

Midterm 2

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Unit 4 1. a) into page b) to the right c) out of page

2. a)	Switch	Coil 1	Coil 2	Coil 3
	Open	Clockwise	Clockwise	Clockwise
	Closed	Counterclockwise	Counterclockwise	Counterclockwise
	Left Closed	No change	No change	No change

3. a) 2 T field 2.20 cm diameter .25 sec

$$A = \pi r^2 = \frac{\pi (0.0220)^2}{4} = .00038 \text{ m}^2 \quad A = \pi (0.011)^2 = 3.8 \times 10^{-4} \text{ m}^2$$

$$\Delta \Phi = B \Delta A = 2 (3.8 \times 10^{-4}) = 7.6 \times 10^{-4} \text{ Wb}$$

$$\text{emf}_{\text{avg}} = \frac{\Delta \Phi}{\Delta t} = \frac{7.6 \times 10^{-4}}{0.25} = 3.04 \times 10^{-3} \text{ V}$$

$$b) I = \frac{\text{emf}}{R} = \frac{3.04 \times 10^{-3}}{0.01} = 304 \text{ A}$$

$$c) P = I^2 R = (304)^2 \cdot 0.01 = 9.24 \times 10^{-4} \text{ W}$$

4. a) 1875 rad/s 18 V peak emf .01m 3cm 1cm .64 T field

$$V_p = NAB\omega \quad N = \frac{V_p}{AB\omega} = \frac{18}{(3 \times 10^{-2})(1)(1875)} = 50 \text{ turns}$$

$$b) W = 2\pi f \quad f = \frac{\omega}{2\pi} = \frac{1875}{2\pi} = 298.4 \text{ Hz}$$

$$T = \frac{1}{298.4 \text{ Hz}} = 3.35 \times 10^{-3} \text{ sec}$$

$$5. a) \frac{240}{120} = \frac{2}{1} = 2$$

$$b) \frac{120}{240} = \frac{1}{2}$$

$$6. c) V_0 = 120 \text{ V} \quad \omega = (240\pi) \text{ Hz} \quad V(t) = 0$$

$$V_t = 120 \sin(240\pi t) \quad V_t = \frac{1}{120} \text{ sec}$$

7. .1 A 2 mH 500V

$$\text{emf} = L \frac{\Delta I}{\Delta t} \rightarrow \Delta t = L \frac{\Delta I}{\text{emf}} = \frac{2 \times 10^{-3} (.1)}{500} = 4 \times 10^{-7} \text{ s}$$

8. a) 25 H = L

$$\text{emf} = -L \frac{\Delta I}{\Delta t} = -25 \cdot \frac{0-100}{80 \times 10^{-3}} = 31250 \text{ V} (31.25 \text{ kV})$$

$$b) W = \frac{1}{2} L I^2$$

$$= \frac{1}{2} (25 (100^2)) = 125000 \text{ J} (125 \text{ kJ})$$

$$c) P = \frac{W}{t} = \frac{125 \times 10^5}{80 \times 10^{-3}} = 1562000 \text{ W} (1.56 \text{ MW})$$

$$d) L = 25 \text{ H} \quad l = .5 \text{ m} \quad A = .01 \text{ m}^2$$

$$L = \frac{\mu_0 N^2 A}{l} = 25 = \frac{(4\pi \times 10^{-7}) N^2 (.01 \text{ m}^2)}{.5 \text{ m}} \Rightarrow N = \frac{25 \cdot .5}{(4\pi \times 10^{-7}) \cdot .01} = 31539.15$$

$$e) X_L = 2\pi f L = 2\pi \times 10^4 \cdot 25 = 1.57 \times 10^6 \Omega$$

9. a) 20 nF $5 \text{ M}\Omega$

$$\tau = RC = L = R \cdot \tau = (5 \times 10^6) 20 \times 10^{-9} = .1 \text{ s}$$

b) $R = \frac{L}{\tau} = \frac{1}{1 \times 10^{-8}} = 1 \times 10^8 \Omega$

c) $I(t) = I_0 e^{-\frac{t}{\tau}} = e^{-\frac{t}{\tau}} = \frac{I(t)}{I_0} = e^{-\frac{t}{\tau}} = e^{-1.5} = 86.1\%$

10. a) $V_{in} = IR$

b) $V_{out} = V_L$

c) $\frac{V_{out}}{V_{in}} = \frac{1}{1 + i\omega RC}$

d) $\omega = 2\pi f = 2\pi (100) \frac{1}{2\pi RC} = \frac{100}{RC}$

$$\frac{V_{out}}{V_{in}} = \frac{1}{\sqrt{1 + 100^2}} \approx .01$$

$$\frac{V_{out}}{V_{in}} = \frac{1}{\sqrt{1 + 0.1^2}} = .995$$

11. a) $R = 1 \text{ k}\Omega$ $C = 1 \text{ MF}$ $L = 10 \text{ mH}$

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{10 \times 10^{-3} \text{ H} \times 1 \times 10^{-6} \text{ F}}} = 1592.36 \text{ Hz}$$

b) $Z = \sqrt{R^2 + (2\pi fL - \frac{1}{2\pi fC})^2}$ $f = 1592.36 \text{ Hz}$

$$\sqrt{(1 \times 10^3)^2 + (2\pi \times 1592.36 \times 10 \times 10^{-3} - \frac{1}{2\pi \times 1592.36 \times 10^{-6}})^2} = 995.03 \text{ H}_2$$

12. a) $V_{rms} = 120 \text{ V}$

$R = 1 \text{ k}\Omega$

$C = 1 \text{ MF}$

$L = 10 \text{ mH}$

$$I_{rms} = \frac{V_{rms}}{Z} = \frac{120}{995} = 120.6 \text{ mA}$$

b) $P_{rms} = V_{rms} \cdot I_{rms} \cdot \cos \theta$

$$\theta = \frac{R}{Z} = \frac{100}{995} = .1005$$

$$P_{rms} = 120 \cdot 120.6 \cdot 1005 = 1.45 \text{ W}$$

13. $f_c = 1.4 \text{ MHz}$

$$f_c + f_m = 1.4 \text{ MHz} + 10 \text{ kHz} = 1.41 \text{ MHz}$$

$$f_c - f_m = 1.4 \text{ MHz} - 10 \text{ kHz} = 1.39 \text{ MHz}$$

Unit 5 1. a) 1 cm laterally

$$B = \frac{\mu_0 I_d}{2\pi r} = \frac{(4\pi \times 10^{-7} \text{ T}\cdot\text{m/A}) (2 \times 10^{-6} \text{ A})}{2\pi (1 \times 10^{-2} \text{ m})}$$

$$\frac{4 \times 10^{-7} (2 \times 10^{-6})}{2 \times 10^{-2}} = 2 \times 10^{-6} \text{ T} = 2 \mu\text{T}$$

$0 - 200 \text{ nC}$ in 2 sec

$$I_d = \frac{\Delta Q}{\Delta t} = \frac{2 \times 10^{-7} \text{ C}}{2 \times 10^{-6} \text{ s}}$$

$$r = 1 \times 10^{-2} \text{ m}$$

b) Displacement current generates a magnetic field, which it is responsible

2. a) $D = C \cdot \Delta t$

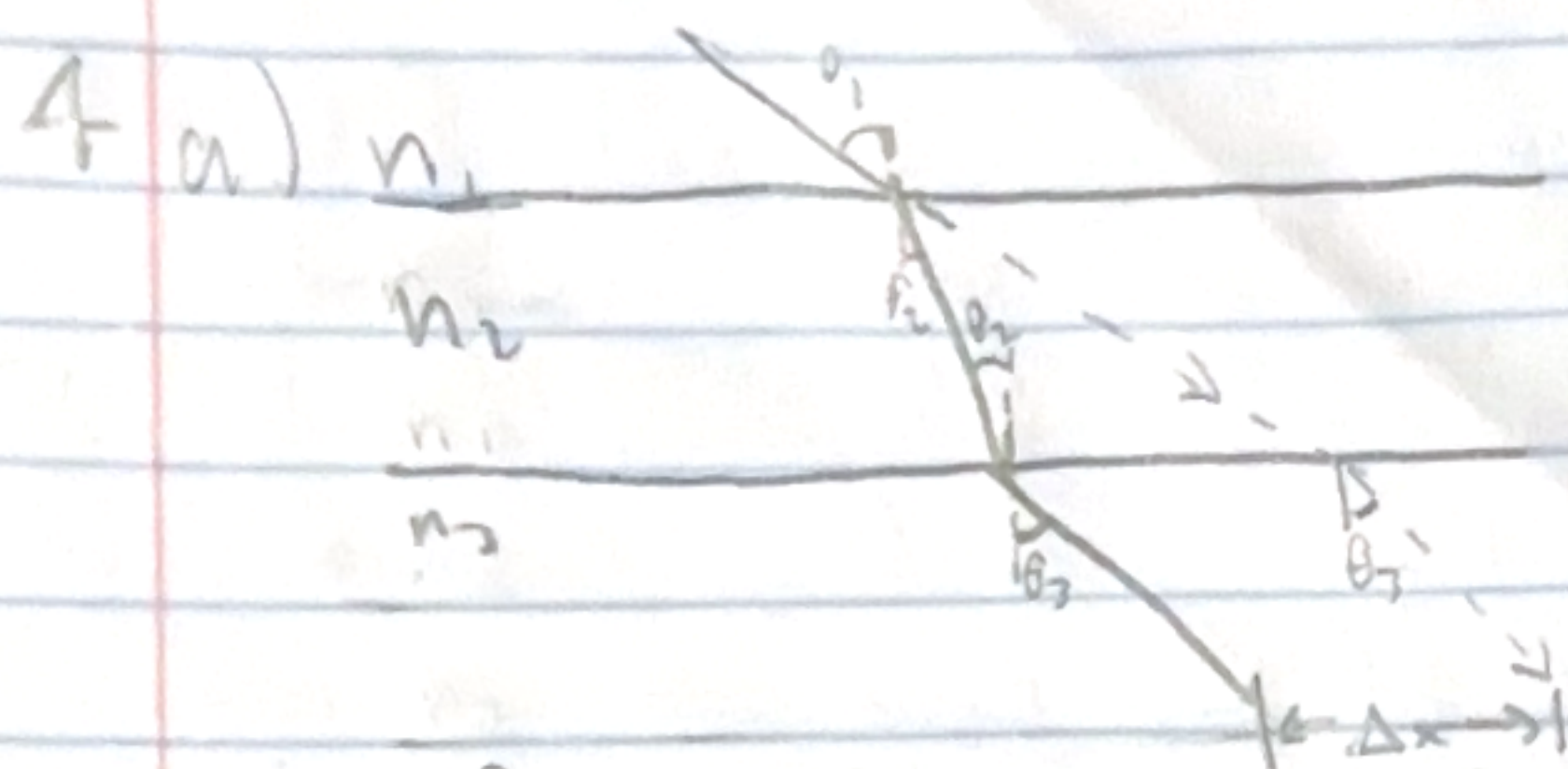
$$C = 3 \times 10^8 \text{ m/s}$$

$$\Delta t = 10 \text{ ns}$$

$$D = 3 \times 10^8 (10 \times 10^{-9} \text{ s}) = 3000 \text{ m round trip} = 1500 \text{ m from aircraft}$$

2. b) $R = \left(\frac{n_1 - n_2}{n_1 + n_2} \right)^2$, $A_{ir} = 1(n.)$, $I_{ir} = 1.78(n)$
 $R = \left(\frac{1 - 1.78}{1 + 1.78} \right)^2 = \left(\frac{-0.78}{2.78} \right)^2 = .2806$, $.0787 = 7.87\%$ of power is reflected

3. a) $I = \frac{P}{A} = \frac{1000W}{10 \times 10^{-4} m^2} = 1 \times 10^6 W/m^2$
 b) $I_2 = I_1 \left(\frac{r_1}{r_2} \right)^2 = 1 \times 10^6 W/m^2 \left(\frac{1m}{2m} \right)^2 = 1 \times 10^6 \left(\frac{1}{4} \right) = 2.5 \times 10^5 W/m^2$
 $r_1 = 1m$, $r_2 = 2m$, $I_1 = 1 \times 10^6 W/m^2$



$\theta_1 = \theta_3$ if $n_1 = n_3$
 $n_1 \sin \theta_1 = n_2 \sin \theta_2$
 $n_2 \sin \theta_2 = n_3 \sin \theta_3$

$n_1 \sin \theta_1 = n_2 \sin \theta_2 = n_3 \sin \theta_3$

b) $M = \frac{d_i}{d_o}$, $\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$, $d = 8$, $\frac{1}{d_o} = \frac{1}{8}$

$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i} \rightarrow \frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_o} = \frac{1}{12} - \frac{1}{8} = \frac{2}{24} - \frac{3}{24} = -\frac{1}{24} = -24 cm$

$M = \frac{24}{8} = 3 \times$ magnification

5. $10keV$, $1cm$

$I = I_0 e^{-\mu x}$, $\mu = 200 \text{ barns}$, $A^2 = 2 \times 10^{-26} m^2$, $x = .01m$

$n = \frac{\rho N_A}{A} = \frac{11.34 \text{ kg/m}^3 (6.022 \times 10^{23})}{207.2 \text{ kg/mol}} = 7.29 \times 10^{28} \text{ atoms/m}^3$

$\frac{I}{I_0} = e^{-3.24 \times 10^{28} \cdot 2 \times 10^{-26} \cdot 1.01} = e^{-6.58} = 1.39 \times 10^{-3} = .139\%$ passes through

6. a) $1 \text{ rad} = \frac{.01 J}{1 \text{ kg}} = .01 \frac{J}{\text{kg}}$
 $125 \text{ rad} = 125 \frac{J}{\text{kg}}$, $\frac{E}{m} = \frac{.25}{60} = .00417 \frac{J}{\text{kg}}$
 $\frac{.00417}{.01} = 0.417 \text{ rad}$

b) $\frac{.25}{2} = .125 \rightarrow \frac{.125}{.01} = 12.5 \text{ rad}$

c) $1 \text{ Gy} = 100 \text{ rad}$, $.125 \times 1 = .125 \text{ Sv}$
 $.125 \text{ Gy} = 12.5 \text{ rad}$
 $RBE = 1$

d) The dose of .125 Sv is enough to cause mild temporary symptoms, but not high enough to cause long term and/or lethal harm